Data Session 9 Key, part 2: What Is a 70% Chance?

84-355 Democracy's Data | Session 9 — Polls, models, and markets | Instructor copy — do not distribute

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All code is given to students. Grade the interpretation.

Where this sits. The back half of Session 9, five days before the country votes. Part 1 took a poll apart; this asks what happens when a poll becomes a probability. It is the last session before the election and it sets up **Nov 5**, where four forecasts get graded at once: the base rate from Oct 20, the poll average, the market, and each student's own written prediction.

Timing matters here. Teach this *before* the election, never after. The whole point is that students commit to a view of what a probability means while the answer is still unknown.

Every number below was run against the data file.

The session in one sentence

A single probability cannot be wrong, so we check 746 of them at once — and the market turns out to be honest where the money is and shaky where it is thin.

The findings

```
c(markets   = nrow(d),
  brier     = round(mean((d$p7 - d$outcome)^2), 4),
  brier_50  = round(mean((0.5 - d$outcome)^2), 4),
  hi_band   = round(mean(d$outcome[d$p7 > 0.9]), 3),
  low_band  = round(mean(d$outcome[d$p7 > 0.1 & d$p7 <= 0.2]), 3))
```

```
#>  markets    brier brier_50  hi_band low_band
#> 746.0000   0.0644   0.2500   0.9780   0.2960
```

- **Brier 0.0644**, against 0.25 for always saying 50% and 0.1566 for the base rate.
- Priced 0.9–1.0 → happened **97.8%**. Priced 0.8–0.9 → **84.6%**. Very good.

- Priced 0.1–0.2 → happened **29.6%**, about double. Marginal: p = 0.037 pooled across 0.1–0.3, on 46 markets, from **10** bands tested.
- **70.6% of markets were priced under 0.10.**

The three things to get said

1. Start with the thing that cannot be done. Ask the room how they would tell whether "72%" was right. Let them flounder for a minute — the honest answer is that they cannot, from one case. **This is the single most useful idea in the session**, and it reframes every post-election argument they have ever heard.

2. Calibration is the answer, and it needs volume. Not "was the forecast right" but "of the things you called 70%, did 70% happen." Then note what that requires: hundreds of resolved questions, which elections do not supply and markets do. **The reason we are using Polymarket is not that it is better than 538 — it is that it leaves a checkable record.**

3. Then make them doubt the interesting result. The low-band miscalibration is exactly the kind of finding a student will want to write up. All 10 bands were tested; one significant result is what chance produces. **Do not let them conclude "markets underprice longshots"** — Option B shows what is actually going on, and it is better.

The line for the board: you cannot grade a forecast, only a forecaster.

Option A — A different moment

Verified output

```
k <- !is.na(d$p30)
c(n = sum(k),
  brier_7d  = round(mean((d$p7[k]  - d$outcome[k])^2), 4),
  brier_30d = round(mean((d$p30[k] - d$outcome[k])^2), 4))
```

```
#>         n  brier_7d brier_30d
#>  677.0000    0.0500    0.0736
```

Brier 0.0500 at seven days, 0.0736 at thirty, on the 677 markets that carry a thirty-day price. The market gets meaningfully better as resolution approaches.

3/4: Reports that the 30-day forecast is worse.

4/4: Says why that is reassuring rather than interesting — information arrives over time, so a forecaster who was *equally* good a month out would be suspicious, not impressive. **A forecast that never improves is not using anything.**

4/4, best: Notices this cuts against Part 4's framing. Seven days out was chosen arbitrarily and flatters the market; the honest way to describe a forecaster is a curve over time, not one number. **Asking "as of when?" is the same "compared to what?" instinct the course has been drilling since Session 4.**

Option B — Only the big markets

Verified output

```r
med <- median(d$volume)
for (lab in c("above", "below")) {
  x <- if (lab == "above") d[d$volume > med, ] else d[d$volume <= med, ]
  cat(sprintf("  %-6s median volume  n=%3d   Brier %.4f\n",
              lab, nrow(x), mean((x$p7 - x$outcome)^2)))
}
```

```
#>   above  median volume  n=373   Brier 0.0516
#>   below  median volume  n=373   Brier 0.0772
```

```r
m <- d[d$p7 > 0.1 & d$p7 < 0.9, ]
cat("\n  mid-range markets only (0.1-0.9):\n")
```

```
#> 
#>   mid-range markets only (0.1-0.9):
```

```r
for (lab in c("above", "below")) {
  x <- if (lab == "above") m[m$volume > median(m$volume), ] else m[m$volume <= median(m$volume), ]
  cat(sprintf("  %-6s n=%3d   priced %.3f   happened %.3f\n",
              lab, nrow(x), mean(x$p7), mean(x$outcome)))
}
```

```
#>   above  n= 87   priced 0.491   happened 0.494
#>   below  n= 87   priced 0.464   happened 0.552
```

This is the best option in the lab and it resolves Part 4.

Overall: high-volume Brier **0.0516**, low-volume **0.0772**. More money, better prices.

Restricted to the mid-range where calibration actually bites: **high-volume markets priced 0.491 and happened 0.494 — near perfect. Low-volume markets priced 0.464 and happened 0.552 — off by 8.7 points.**

3/4: Reports that busier markets score better.

4/4: Connects it to Part 4 — the apparent longshot bias is concentrated in thinly traded markets, so "markets underprice longshots" is the wrong lesson. **"Thin markets are noisy" is the right one**, and it is what anybody who has thought about prices would predict.

4/4, best: Draws the general moral: the market is not one forecaster but hundreds of forecasters of wildly varying quality, pooled by us into a single table. **We manufactured the anomaly by averaging over things that should not have been averaged** — which is the ecological-inference problem from a different direction, and worth naming if Session 14 is on the horizon.

Option C — Score yourself

No verified output; grade the honesty and the method.

3/4: Produces five probabilities and a Brier score.

4/4: Wrote the numbers down *before* looking, says so, and notices that five questions is far too few to distinguish their skill from luck — the same sample-size problem as the class survey in Session 8.

4/4, best: Notices they will be tempted to pick questions they already know the answer to, and describes how they avoided it. **Recognising your own selection effect is the skill this course is actually teaching.**

Grading

Four-point scale from the syllabus.

- **Never penalise code.** It is your code.
- **The move that earns a 4** is treating a significant result as a hypothesis rather than a conclusion — the multiple-comparisons point in Part 4.
- **A 2** is "prediction markets are accurate" or "prediction markets are biased." Both overclaim from the same table.
- **Reward anyone who asks what the forecasters themselves publish.** The answer — that model outputs are not downloadable — is a genuine finding about the industry.

Anticipated questions

- *"Isn't betting on elections illegal?"* — Kalshi is CFTC-regulated and offers election contracts; Polymarket's US status has changed more than once. Say the legal position is contested and moving, and do not assert a current answer you have not checked that week.
- *"Why not use 538?"* — Because you cannot check it. **No downloadable file of model probabilities exists**, so there is nothing to score. Worth dwelling on: the most-quoted forecasts in America publish claims, not evidence.
- *"Do markets beat models?"* — Unanswerable from this data, and be firm about that. We have one side of the comparison.
- *"Isn't this just gambling?"* — Yes, and the fact that it produces well-calibrated probabilities is exactly what makes it interesting.
- *"Are these political markets?"* — Not all. Sport, crypto and Fed decisions are in the sample. We are testing the mechanism, not the electorate.

Connections

- **Session 9 part 1 (polls)** — a share is not a probability; this is the bridge.
- **Oct 20 (the midterm base rate)** — students wrote a prediction. **Remind them today that it will be scored on Nov 5.**
- **Session 10 (Nov 5)** — four forecasts get graded: base rate, poll average, market, and their own. **This session is the one that tells them what "graded" can even mean.**
- **Session 5 (ads)** — the other place where the interesting evidence is not published.